

ECON 6018

TUTORIAL 3

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- Lesson Tutorial
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Footnote: The practice questions are adapted from Essential Mathematics for Economic Analysis (5th Edition) by Knut Sydsæter, Peter Hammond, Arne Strøm, and Andres Corguel. All credit for the original questions belongs to the authors. The solutions and explanations provided here are my own work.

Differentiation

- the most important thing is your foundation and speed for comparative statics
- For a function to be differentiable, it has to be continuous
- welcome to my favorite Topic!

Chapt 6

Chapt 7

Chapt 11

Chapt 12

1. Derivatives and how we derive them using limits

Derivative — rate of change

- how much y change when x changes
- formula:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- Application
maximization of utility

Questions:

Q1. $f(x) = 4x^2$

Show $f(5+h) - f(5) = 40h + 4h^2$

Find $f'(5)$

Q2. $f(x) = 3x^2 + 2x - 1$

$$\frac{f(x+h) - f(x)}{h} = 6x + 2 + 3h$$

Find $f'(x)$

Answers:

$$\text{Q1. } f(x) = 4x^2$$

$$f(5+h) - f(5) = 40h + 4h^2 \text{ (shown)}$$

$$\lim_{h \rightarrow 0} \frac{40h + 4h^2}{h} = 40$$

$$f'(5) = 40$$

$$\text{Q2. } f(x+h) = 3x^3 + 6xh + 3h^2 + 2x + 2h - 1$$

$$\frac{f(x+h) - f(x)}{h} = 6x + 3h + 2$$

$$h \rightarrow 0$$

$$f'(x) = 6x + 2$$

2. Differentiation

2.1 Basic Differentiation — chpt 6.6

$$f(x) = x^a$$

$$f'(x) = ax^{a-1}$$

Questions

Find the following derivatives:

1(a) $y = 5$ (b) $y = x^4$ (c) $y = 9x^{10}$ (d) $y = \pi^7$

2(a) $2g(x) + 3$ (b) $-\frac{1}{6}g(x) + 8$ (c) $\frac{g(x) - 5}{3}$

3(a) x^6 (b) $3x''$ (c) x^{50} (d) $-4x^{-7}$ (e) $\frac{x^{12}}{12}$ (f) $\frac{-2}{x^2}$ (g) $\frac{3}{\sqrt{x}}$
(h) $\frac{-2}{x\sqrt{x}}$

$$1(a) 0$$

$$(b) 4x^3$$

$$(c) 90x^9$$

$$(d) 0$$

$$2(a) 2g'(x)$$

$$(b) -\frac{1}{6}g'(x)$$

$$(c) \frac{g'(x)}{3}$$

$$3(a) 6x^5$$

$$(b) 33x^{10}$$

$$(c) 50x^{49}$$

$$(d) 28x^{-8}$$

$$(e) x^{11}$$

$$(f) -4x^{-3}$$

$$(g) -x^{-4/3}$$

$$(h) 3x^{-5/2}$$

2.2 Sum, Product, Quotient — chpt 6.7

a. Sum

$$F(x) = f(x) \pm g(x) \Rightarrow F'(x) = f'(x) \pm g'(x)$$

b. Product

$$F(x) = f(x) \cdot g(x) \Rightarrow F'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

c. Quotient

$$F(x) = \frac{f(x)}{g(x)} \quad F'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

Questions:

Differentiate w.r.t x

1 (a) $x+1$ (b) $x+x^2$ (c) $3x^5+2x^4+5$ (d) $8x^4+2\sqrt{x}$

(e) $\frac{1}{2}x - \frac{3}{2}x^2 + 5x^3$ (f) $1-3x^7$

2 (a) $\frac{3}{5}x^2 - 2x^7 + \frac{1}{8} - \sqrt{x}$ (b) $(2x^2-1)(x^4-1)$

(c) $(x^5 + \frac{1}{x})(x^5+1)$

3 (a) $\frac{\sqrt{x}-2}{\sqrt{x}+1}$ (b) $\frac{x^2-1}{x^2+1}$ (c) $\frac{x^2+x+1}{x^2-x+1}$

$$1(a) \ 1 \quad (b) \ 1+2x \quad (c) \ 15x^4 + 8x^3$$

$$(d) \ 32x^3 + 2x^{-\frac{1}{2}} \quad (e) \ \frac{1}{2} - 3x + 15x^2$$

$$(f) -2bc^6$$

$$2(a) \ \frac{6}{5}x - 14x^6 - \frac{1}{2}x^{-\frac{1}{2}}$$

$$(b) \ (4x^3)(x^4-1) + (2x^2-1)(4x^3)$$

$$(c) \ (5x^4 + x^{-2})(x^5+1) + (2x^5 + \frac{1}{x})(5x^4)$$

$$3(a) \ (\frac{1}{2}x^{-\frac{1}{2}})\left(\frac{1}{\sqrt{x}+1}\right) + (\sqrt{x}-2)(\sqrt{x}+1)^{-2}\left(\frac{1}{2}x^{-\frac{1}{2}}\right)$$

$$(b) \ \frac{(2x)(x^2+1) - (x^2-1)(2x)}{(x^2+1)^2}$$

$$(c) \ \frac{(2x+1)(x^2-x+1) - (x^2+x+1)(2x-1)}{(x^2-x+1)^2}$$

2.3 Chain Rule ★ — chpt 6.8

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Questions :

1(a) $y = 5u^4$

where $u = 1 + x^2$

(b) $y = u - u^6$

where $u = 1 + \frac{1}{x}$

1. Use the chain rule (6.8.1) to find dy/dx for the following:

(a) $y = 5u^4$, where $u = 1 + x^2$

(b) $y = u - u^6$, where $u = 1 + 1/x$

2. Compute the following:

(a) dY/dt , when $Y = -3(V+1)^5$ and $V = \frac{1}{3}t^3$.

(b) dK/dt , when $K = AL^a$ and $L = bt + c$, where A, a, b , and c are positive constants.

2.(a) Find $\frac{dY}{dt}$, $Y = -3(V+1)^5$ and $V = \frac{1}{3}t^3$

(b) Find $\frac{dK}{dt}$, $K = AL^a$ and $L = bt + c$ where A, a, b, c are +ve constants

Answer:

$$1(a) \frac{dy}{dx} = 20u^3 (2x)$$

$$= 20(1+x^2)^3(2x)$$

$$(b) \frac{dy}{dx} = (1-6u^5)(-x^{-2})$$

$$= (1-1(1+\frac{1}{x})^5)(-x^{-2})$$

$$2(a) \frac{dy}{dt} = [-15(1+t^3)^4](t^2)$$

$$= [-15(\frac{1}{3}t^3+1)^4](t^2)$$

2.4 Higher Order Derivatives — Chpt 6.09

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

Question: Find the second derivative

(a) $y = x^5 - 3x^4 + 2$

(b) $y = (1 + x^2)^{10}$

Ans:

$$(a) \frac{dy}{dx} = 5x^4 - 12x^3$$

$$\frac{d^2y}{dx^2} = 20x^3 - 36x^2$$

$$(b) \frac{dy}{dx} = 10(1+x^2)^9 \cdot 2x \\ = 20x (1+x^2)^9$$

$$\frac{d^2y}{dx^2} = 20(1+x^2)^9 + 20x \cdot 9(1+x^2)^8 \cdot 2x$$

2.5 Exponential — Chpt 6.10

$$f(x) = e^{g(x)} \quad f'(x) = g'(x)e^{g(x)}$$

Questions:

Find first and second derivatives

(a) $y = e^{-3x}$ (b) $y = 2e^{x^3}$ (c) $y = e^{\frac{1}{x}}$ (d) $y = 5e^{2x^2-3x+1}$

Find the first and second derivatives of:

(a) $y = e^{-3x}$ (b) $y = 2e^{x^3}$ (c) $y = e^{1/x}$ (d) $y = 5e^{2x^2-3x+1}$

Ans:

$$(a) -3e^{-3x}$$

$$(b) 6x^2 e^{x^3}$$

$$(c) -x^2 e^{\frac{1}{x}}$$

$$(d) (4x+3) \cdot 5e^{2x^2-3x+1}$$

2.6 Logarithmic — Chpt 6.11

$$y = \ln h(x)$$

$$y' = \frac{h'(x)}{h(x)}$$

Questions

Find derivatives

(a) $y = x^3 (\ln x)^2$ (b) $y = \frac{x^2}{\ln x}$ (c) $y = (\ln x)^{10}$

(d) $y = (\ln x + 3x)^2$

Ans:

$$(a) 3x^2 (\ln x^2) + (3x^2) (2 \ln x) \left(\frac{1}{x}\right)$$

$$(b) \frac{2x \cdot \ln x - \frac{1}{x} \cdot x^2}{(\ln x)^2}$$

$$(c) 10 \cdot (\ln x)^9 \cdot \frac{1}{x}$$

$$(d) 2(\ln x + 3x) \cdot \left(\frac{1}{x} + 3\right)$$